

Recall:

If  $A = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix}$  is an  $m \times n$  matrix then:

- 1)  $\text{Col}(A) = \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_n)$
- 2)  $\text{Nul}(A) = \{\mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \mathbf{0}\}$

## Construction of a basis of $\text{Col}(A)$

### Lemma

Let  $V$  be a vector space, and let  $\mathbf{v}_1, \dots, \mathbf{v}_p \in V$ . If a vector  $\mathbf{v}_i$  is a linear combination of the other vectors then

$$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p) = \text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_p)$$

**Upshot.** One can construct a basis of a vector space  $V$  as follows:

- Start with a set of vectors  $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$  such that  $\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_p) = V$ .
- Keep removing vectors without changing the span, until you get a linearly independent set.

**Example.** Find a basis of  $\text{Col}(A)$  where  $A$  is the following matrix:

$$A = \begin{bmatrix} 1 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

**Example.** Find a basis of  $\text{Col}(A)$  where  $A$  is the following matrix:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

### Construction of a basis of $\text{Nul}(A)$

**Example.** Find a basis of  $\text{Nul}(A)$  where  $A$  is the following matrix:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

**Upshot.** If  $A$  is matrix then:

$\dim \text{Col}(A) =$  the number of pivot columns of  $A$

$\dim \text{Nul}(A) =$  the number of non-pivot columns of  $A$

### Definition

If  $A$  is a matrix then:

- the dimension of  $\text{Col}(A)$  is called the *rank* of  $A$  and it is denoted  $\text{rank}(A)$
- the dimension of  $\text{Nul}(A)$  is called the *nullity* of  $A$ .

### The Rank Theorem

If  $A$  is an  $m \times n$  matrix then

$$\text{rank}(A) + \dim \text{Nul}(A) = n$$

## Low rank matrices in data analysis

|           | <i>exam 1</i> | <i>exam 2</i> | <i>exam 3</i> | <i>exam 4</i> |
|-----------|---------------|---------------|---------------|---------------|
| student 1 | 2             | 1             | 3             | 10            |
| student 2 | 4             | 2             | 6             | 20            |
| student 3 | 10            | 5             | 15            | 50            |
| student 4 | 2             | 1             | 3             | 10            |
| student 5 | 0             | 0             | 0             | 0             |
| student 6 | 6             | 3             | 9             | 40            |